Enrollment No.: 240840116065

Practical-8

AIM- Create interactive visualizations using Plotly Express and Plotly Graph Objects.

Step 1: Import Required Libraries

```
# Import Pandas.
import pandas as pd
 # Import Plotly Express.
import plotly.express as px
 # Import Ploty Graph Objects.
import plotly.graph_objects as go
```

Step 2: Create Dataset

```
data = {
    "Month": ["January", "February", "March", "April", "May", "June",
              "July", "August", "September", "October", "November", "December"],

    "Product": ["Laptop", "Mobile", "Tablet", "Laptop", "Mobile", "Tablet",
                "Laptop", "Mobile", "Tablet", "Laptop", "Mobile", "Tablet"],

    "Region": ["North", "South", "East", "West", "North", "South",
               "East", "West", "North", "South", "East", "West"],

    "Sales": [120000, 135000, 150000, 145000, 160000, 175000,
              180000, 170000, 155000, 190000, 200000, 210000],

    "Units_Sold": [12, 30, 25, 14, 35, 30, 18, 38, 32, 21, 45, 40],

    "Profit": [18000, 27000, 30000, 22000, 32000, 35000,
               28000, 34000, 38000, 40000, 42000, 45000]
}

df = pd.DataFrame(data)
df
```

	Month	Product	Region	Sales	Units_Sold	Profit
0	January	Laptop	North	120000	12	18000
1	February	Mobile	South	135000	30	27000
2	March	Tablet	East	150000	25	30000
3	April	Laptop	West	145000	14	22000
4	May	Mobile	North	160000	35	32000
5	June	Tablet	South	175000	30	35000
6	July	Laptop	East	180000	18	28000
7	August	Mobile	West	170000	38	34000
8	September	Tablet	North	155000	32	38000
9	October	Laptop	South	190000	21	40000
10	November	Mobile	East	200000	45	42000
11	December	Tablet	West	210000	40	45000

Step 3: Inspect the Dataset

```
# Display First Five Columns.
df.head()
```

	Month	Product	Region	Sales	Units_Sold	Profit
0	January	Laptop	North	120000	12	18000
1	February	Mobile	South	135000	30	27000
2	March	Tablet	East	150000	25	30000
3	April	Laptop	West	145000	14	22000
4	May	Mobile	North	160000	35	32000

```
# Display No. of Rows and Columns.
df.shape
```

```
(12, 6)
```

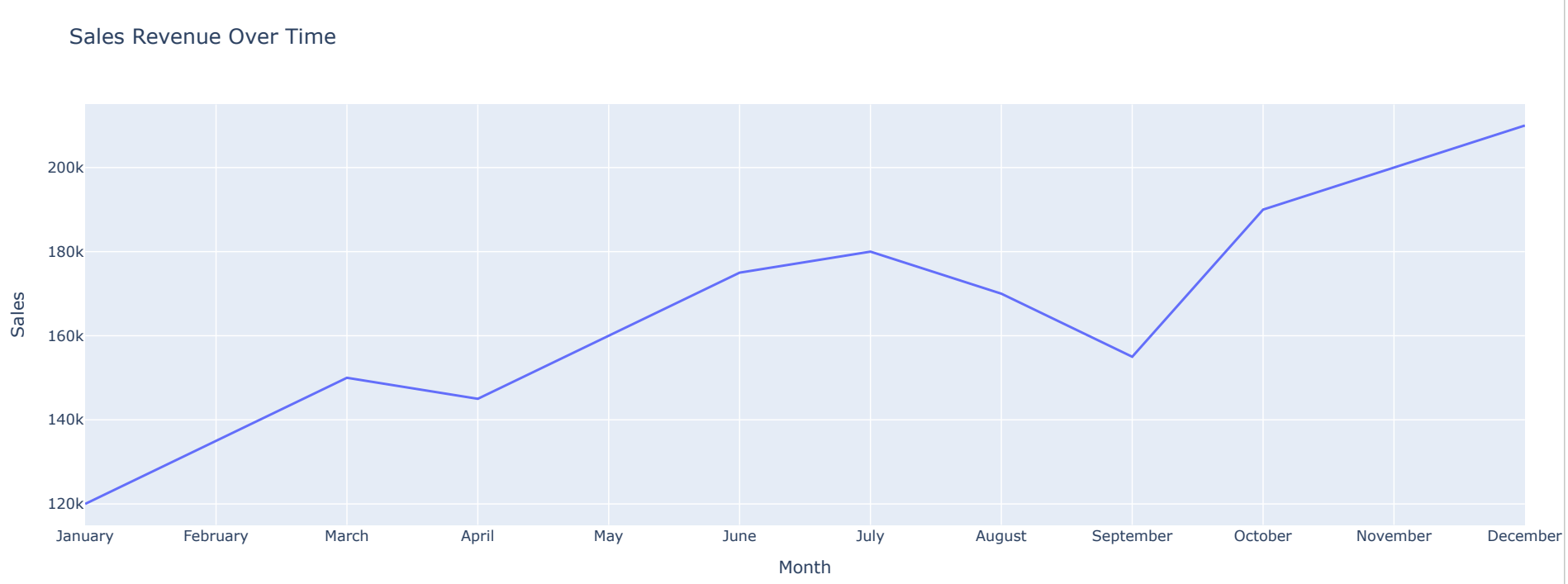
```
# Display Information od dataset.
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 12 entries, 0 to 11
```

```
Data columns (total 6 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   Month       12 non-null     object
 1   Product     12 non-null     object
 2   Region      12 non-null     object
 3   Sales       12 non-null     int64
 4   Units_Sold  12 non-null     int64
 5   Profit      12 non-null     int64
dtypes: int64(3), object(3)
memory usage: 708.0+ bytes
```

Step 4: Create an Interactive Line chart using Plotly Explore

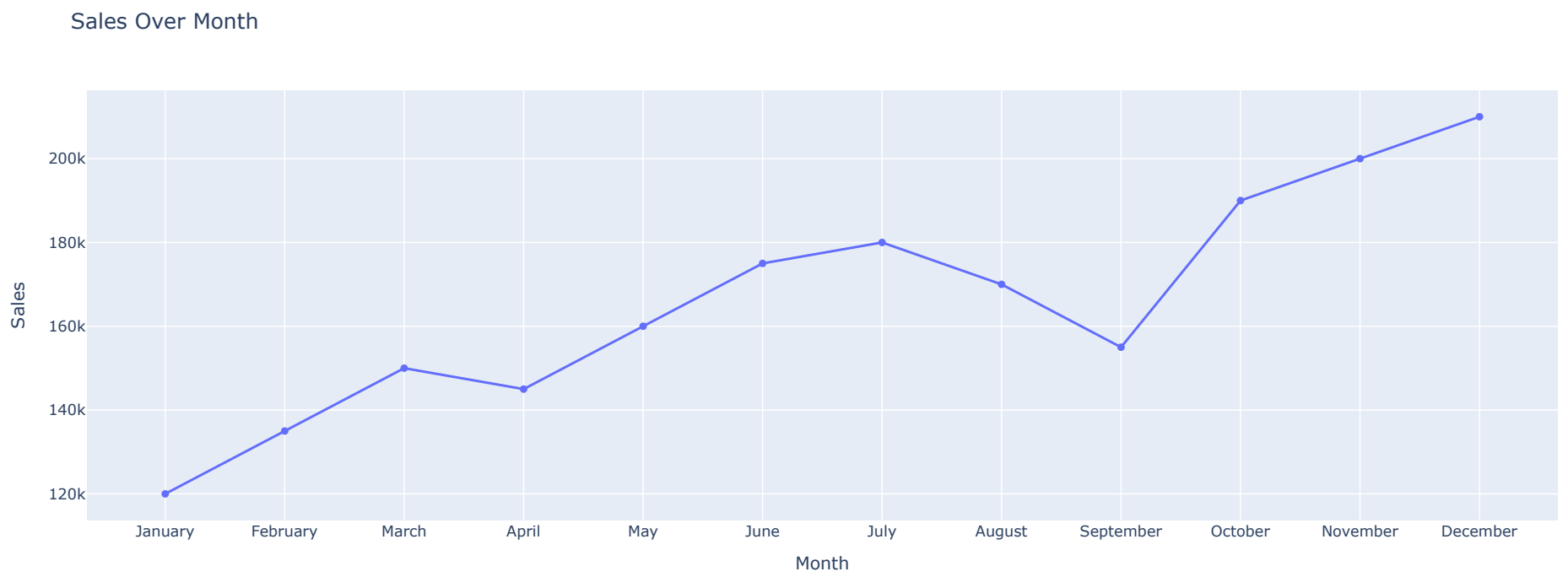
```
# How does the sales revenue change throughout the year?
fig = px.line(df, x="Month", y="Sales", title="Sales Revenue Over Time")
fig.show()
```



Step 5: Improve the Hover Information

```
fig = px.line(
    df,
    x='Month',
    y='Sales',
    title='Sales Over Month',
    markers=True
)

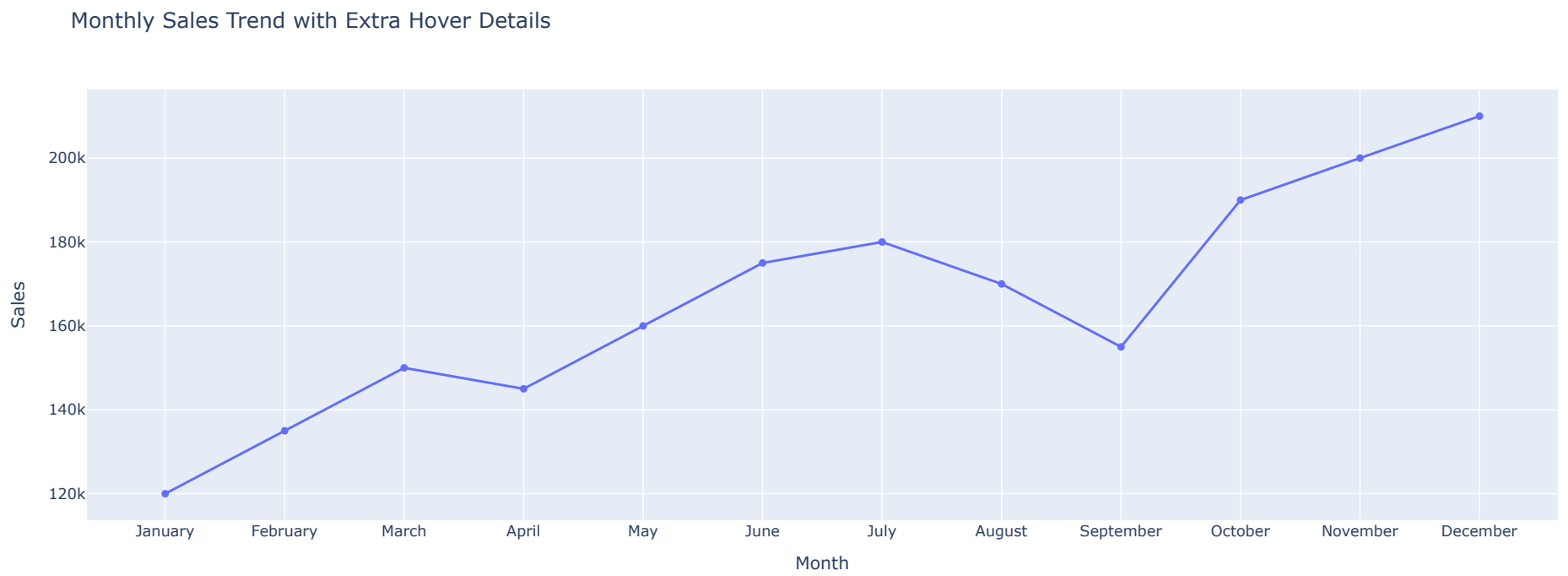
fig.show()
```



```
import plotly.express as px

fig = px.line(
    df,
    x="Month",
    y="Sales",
    markers=True,
    title="Monthly Sales Trend with Extra Hover Details",
    hover_data=["Product", "Region", "Units_Sold", "Profit"]
)

fig.show()
```



Step 6: Create an Interective Bar Chart

```
# Group the dataset according to Product
product_sales = df.groupby('Product')['Sales'].sum().reset_index()

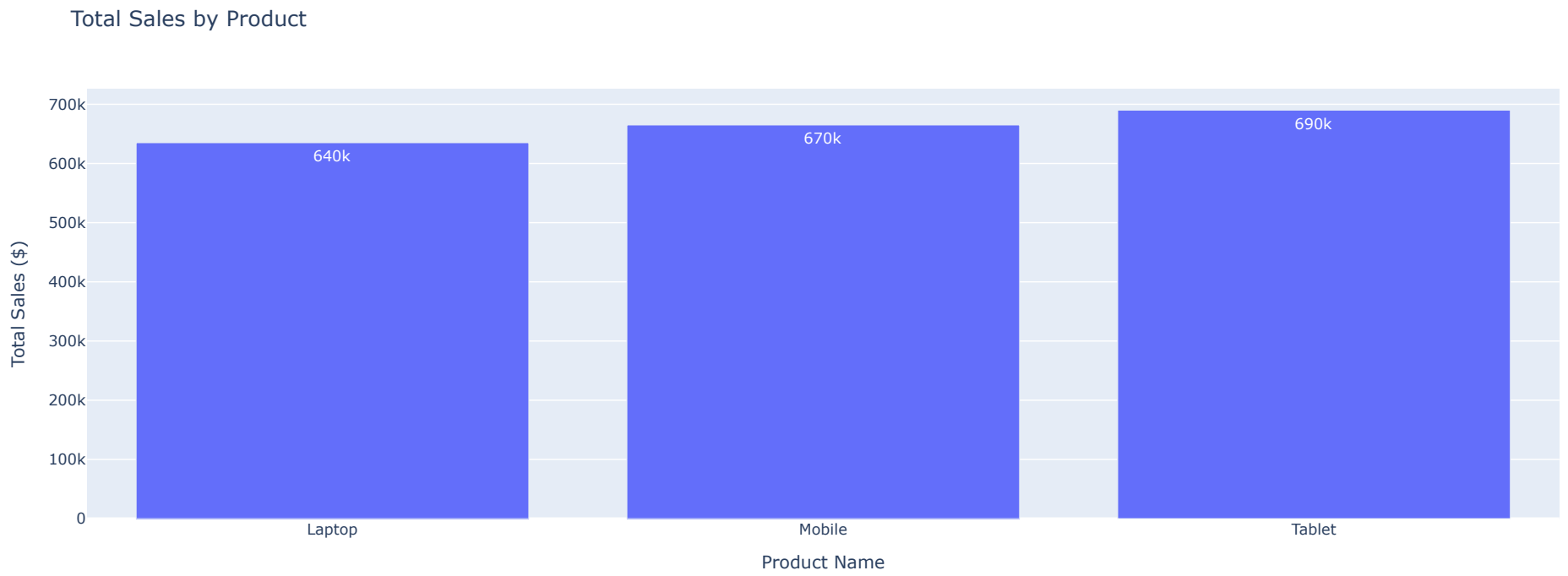
# Calculate total Sales generated by each product
print(product_sales)
```

```
  Product   Sales
0  Laptop  635000
1  Mobile  665000
2  Tablet  690000
```

```
# Create an interactive bar chart

fig = px.bar(
    product_sales,
    x='Product',
    y='Sales',
    title='Total Sales by Product',
    labels={
        'Sales': 'Total Sales ($)',
        'Product': 'Product Name'
    },
    text_auto='.2s'
)

fig.show()
```

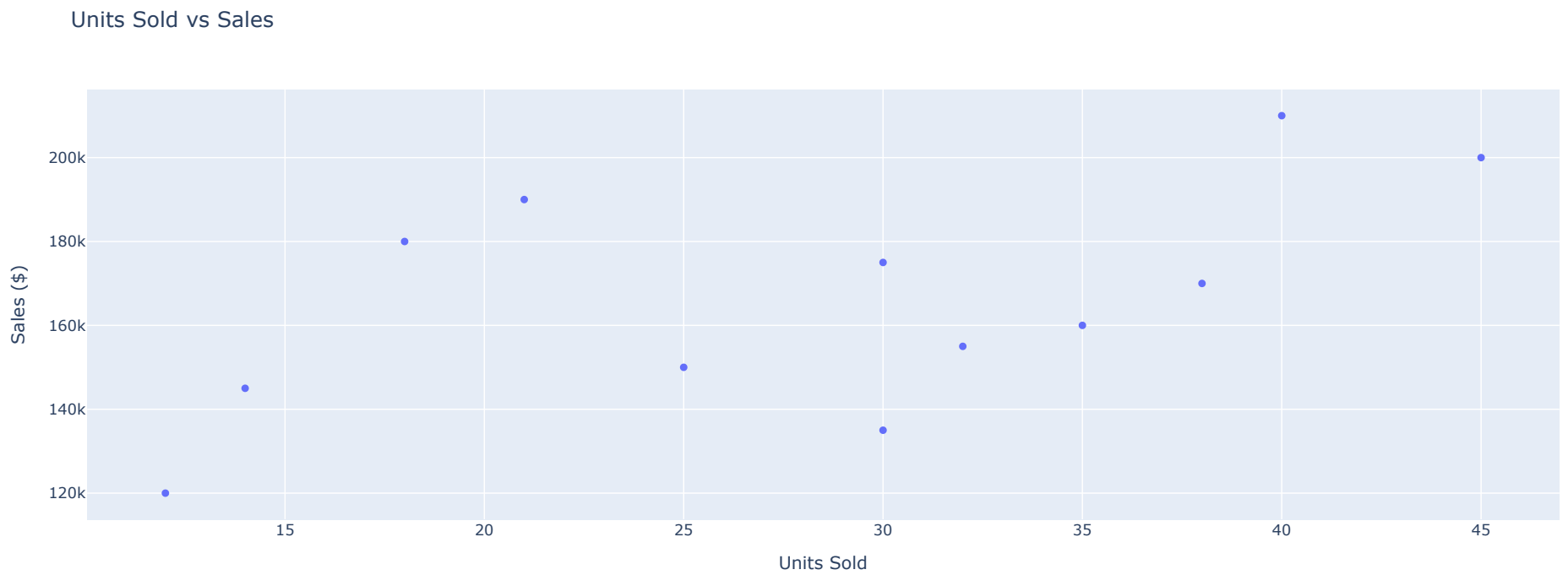


Step 7: Interective Scatter Plot

```
# Analyze relationship between Units Sold and Sales

fig = px.scatter(
    df,
    x="Units_Sold",
    y="Sales",
    title="Units Sold vs Sales",
    labels={
        "Units_Sold": "Units Sold",
        "Sales": "Sales ($)"
    }
)

fig.show()
```

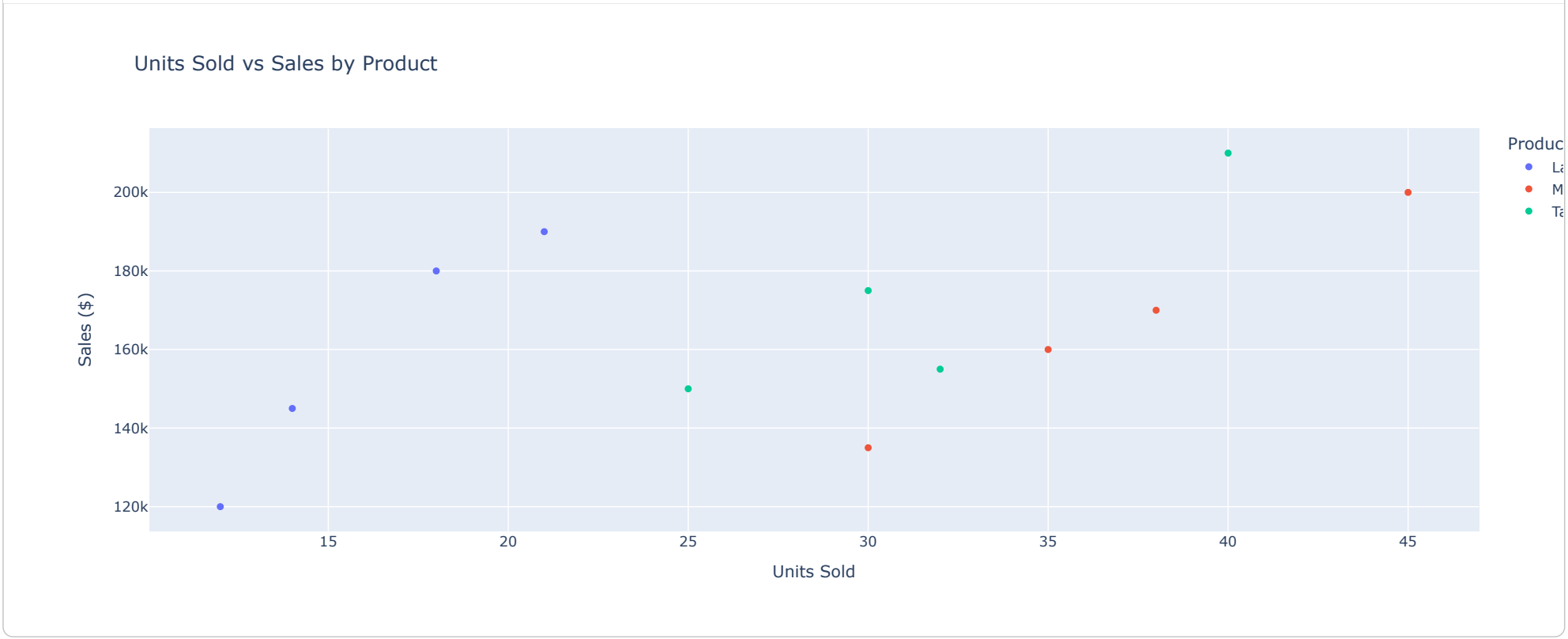


Step 8: Interective Scatter Plot with Color

```
# Create an interactive scatter plot with color

fig = px.scatter(
    df,
    x="Units_Sold",
    y="Sales",
    color="Product",
    title="Units Sold vs Sales by Product",
    labels={
        "Units_Sold": "Units Sold",
        "Sales": "Sales ($)"
    }
)

fig.show()
```



Step 9: Save thye Dataset

```
# Save dataset in CSV file

df.to_csv("DAta_Analysis.csv", index=False)
```

```
print("CSV file saved Successfully")

from google.colab import files

files.download("DAta_Analysis.csv")
```

```
CSV file saved Successfully
```